



TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY
Session 202309

BMCS2123 NATURAL LANGUAGE PROCESSING



Choong Zhi Yang
2213552
VADER technique

Title	:	Sentiment Analysis on Out-Of-Vocabulary (OOV) Malaysia Rojak Language
Tutorial Class	:	RDS Y2 S1 G1
Tutor Name	:	NOOR AIDA BINTI HUSAINI
Submission Date	:	25/12/2023

Table of Contents

1.0 Introduction.....	2
2.0 Research Background.....	5
3.0 Methodology.....	8
4.0 Results.....	27
5.0 Discussion and Conclusion.....	33
6.0 References.....	36

1.0 Introduction

1.1 Existing Problems

The comment or review of a product or services for a company gives a wide range of information about customer views and opinions. This information is valuable for company management, as it helps in marketing, managing relationships with customers, and retaining customers (Understanding Customer Experience, n.d.). Sentiment analysis system trained using NLP is one of the tools to interpret and understand the customer's emotion or views. Although sentiment analysis has been shown to be helpful in analyzing customer sentiment, marketing campaigns, and product reviews; but, when applied to the Malaysian Rojak language, it adds a new level of complexity and significance.

In the context of Malaysian Rojak Language, which is a unique grammatical fusion, sentiment analysis becomes a complex and difficult task. In particular, people in Malaysia use the Malay language as the primary language, followed by English and Chinese. Often, Malaysians mix Malay and English, creating a unique Malaysia mixed language or rojak language (BRCC and SentiBahasaRojak: The First Bahasa Rojak Corpus for Pretraining and Sentiment Analysis Dataset, 2022). Unlike standardized languages, the rojak language provides a special problem due to its combination of Malay and English. It needs specialized methods for language preprocessing and sentiment categorization to fully understand and interpret the sentiments in the sentences.

For e-commerce companies operating in Malaysia, where the Rojak language is often used, sentiment analysis becomes a crucial tool. Advanced language processing methods are required due to different language complexities. For e-commerce platforms to obtain accurate information into customers' views and emotions about products and services, it is important that they analyze the customer sentiments in the original comment. Advanced natural language processing (NLP) methods are required due to the complexity of mixed-language words, even though the unstructured information from customer feedback in Rojak language provides rich insights. The

difficulty is in developing systems that can accurately interpret sentiments in addition to understanding the differences in language.

1.2 Aim

Most of the sentiment analysis in the market focuses on expressions in a single language, especially in English, and less on multilingual expressions that mix more than two languages. Thus, the goal of this project is to fill this gap by creating a sentiment analysis technique that can work in Malaysia's multilingual or mixed-language environment. We aim to design and implement a sentiment analysis system that can accurately analyze the sentiment of text written in out-of-vocabulary (OOV) Malaysia Rojak language.

1.3 Objective

The project's first objective is to fill the gap in sentiment analysis techniques in mixed language environments like Malaysian Rojak language. The project improves and refines the existing sentiment analysis methodologies, which focus on single-language methods. By improving the sentiment analysis techniques for the complex mix-language structure, the project was able to make a breakthrough in this field.

The next objective is to provide a trustable and accurate sentiment analysis system for e-marketing companies that operate in Malaysia. Given that most of the sentiment analysis on the market are designed for the single language, there is a distinct lack of solution for the complexity of the Malaysian Rojak language. This project aims to provide the e-marketing field in Malaysia with a robust system that is capable of not only understanding but also can accurately interpret emotions expressed in a unique language that mixes Malay and English.

The deployed sentiment analysis system is expected to develop into an accurate tool that provides e-marketing companies with a deeper understanding of customer opinions expressed for their service and product in the Rojak language in Malaysia. Making correct and intelligent choices in company business, such as marketing strategies, new product projects, and customer service areas. The ultimate objective is to enable these companies to accurately navigate Malaysia's complex language environment, allowing more accurate and strategic decision-making processes. By doing this, the project hopes to increase

sentiment analysis environment in multi language while also improving the effectiveness and success of e-marketing companies in Malaysia's multilingual background.

1.4 Solution

To handle the complexity of Malaysian Rojak language, we focus more on the text preprocessing part. Before that, we are scrapping the dataset from the shopee product comment, which has both english and malay languages or a mix of them. In handling of text preprocessing, this project uses creative methods such as emoji to text translation, malay word short form normalization, malay word translation to english, different text annotation or labeling by using different libraries and using different methods to reduce inflected words to a common base root. After all of this preprocessing is done, 9 datasets with different groups of combinations of methods used in various steps are created for further model training. The reason for using different groups of methods for preprocessing is to compare and find the best group of data preprocessing methods. After preprocessing, we investigate various machine learning models, including decision tree, random forest, SVM, KNN, logistic regression, and naive bayes, to identify the model with the highest accuracy for deployment. After all the text preprocessing and model training, design and deploy the sentiment analysis system that can accurately analyze the sentiments of out-of-vocabulary Malaysian Rojak language by using streamlit. All of the explanation and reason of methods used will be explained in the 3.0 methodology section.

2.0 Research Background

2.1 Future trend of sentiment analysis in ecommerce.

Based on the research of the article, it shows that the sentiment analysis system is crucial for the ecommerce company. It brings a lot of benefits for the company which include understanding the customer's emotion or views that are helpful in marketing, managing relationships with customers, and retaining customers for the ecommerce company (Customer Sentiment Analysis: A Crucial Need in E-Commerce, 2015). In addition, based on the research from ModorIntelligence that show that the Malaysia E-commerce Market size is estimated at USD 9.38 billion in 2023, and is expected to reach USD 18.31 billion by 2028, growing at a CAGR of 14.32% during the forecast period (2023-2028) (Malaysia E-Commerce Market - Statistics, Share & Growth, n.d.). This means that the ecommerce industry in Malaysia will expand and more companies will be developed. And most of the research on sentiment analysis focuses on single-language expressions, English in particular, and less on multilingual expressions that mix more than two languages or dialects (A Mixed Malay-English Language COVID-19 Twitter Dataset: A Sentiment Analysis, n.d.). So, this means that there is a distinct lack of sentiment analysis system in the market for the ecommerce company in Malaysia to analyze the Malaysian customer's emotion due to the complexity of the Malaysian Rojak language. So, there is a clear need for an advanced sentiment analysis system that can accurately predict the sentiment of the Malaysian rojak language.

2.2 Translation used

For the technique used in the project, most of the sentiment analysis projects have the same technique used. For mixed language, the technique is slightly different, one of the methods is to translate all the text into one language and another method is to train both language models. From a paper that performed machine translations in Arabic tweets to English, it showed that both strategies, a translation-based and a native method perform equally well (Benchmarking Machine Translated Sentiment Analysis for Arabic Tweets, 2015). And from a paper that uses the combination of two sentiment analysis methods, the authors translate an original corpus from German, Russian and Spanish to English. Then, the results from the translated text are compared with native methods, where, in the worst case it was only 5% inferior (Shalunts, G., Backfried, G., & Commeignes, N., 2016).

2.3 Automatic annotation vs Manual annotation

When dealing with data annotation, there are 2 ways to handle it. One is the data annotation using the library pretrained, or manually data annotation by human. Based on the research, it is highly recommended to use automatic pre-annotation in syntactic annotation projects as it increases the Quality and Efficiency when doing data annotation (Stepankova, 2023). Next, other research papers also state that manual annotation is a time-consuming process that involves adding metadata to datasets by human annotators. This method allows for greater control and accuracy in data labeling but can also lead to poor quality due to human errors (Dupont, 2023). In addition, papers from tech science press also have the same research result saying that manual annotation is a time consuming, costly and laborious process. To overcome these resource constraints this research has proposed a fully automated annotation technique for aspect level sentiment analysis. It is very efficient and requires very less time without any additional cost. The performance ratio of manual to proposed i.e., 1:11934.28 shows its efficiency. If the complete dataset were annotated manually then it might have taken approximately 2.07 million seconds, 14 weeks, 1 day and 7 h while working 40 h a week (i.e., 575 h) but this technique has done the same with 173.53 s (2 min and 53.45 s) (Tech Science Press, 2019).

2.4 Why no neutral

When dealing with data annotation, some of the research papers show that the label “neutral” is sometimes used as a label for the objective class (“lack of opinion”) in the literature (Opinion Mining and Sentiment Analysis, 2007). In addition, some research paper observe that neutral comments in feedback systems are not necessarily perceived by users as lying at the exact midpoint between positive and negative comments; rather, “the information contained in a neutral rating is perceived by users to be much closer to negative feedback than positive” (THE DYNAMICS OF SELLER REPUTATION: EVIDENCE FROM EBAY, n.d.). In research of Pang and Lee, it observe that a one vs-all multi-class categorization scheme can outperform regression for a three-class classification problem (positive, neutral, and negative), perhaps due to each class exhibiting a sufficiently distinct vocabulary, but for more fine-grained classification, regression emerges as the better of the two (Seeing Stars: Exploiting Class Relationships for Sentiment Categorization With Respect to Rating Scales, n.d.).

2.5 Lemmatization vs Stemming

In text preprocessing, Lemmatization and Stemming play a major role in text and natural language processing. Both of them create the base part of the inflected word-texts. The difference lies in the fact that stem is not a real word-text whereas lemma is a real language text format (BS, n.d.). Another research paper states that there is no difference between lemmatization and stemming for Analysis of Classification. This paper making experiment on whether using lemmatization approaches can improve the recall compared to the stemming approach. The result of this paper shows that lemmatization also failed to be analyzed in the context of medical systematic reviews (Nunzio, n.d.).

3.0 Methodology

Illustration of the system design (advice to redraw, not directly copy paste from internet)

Discussion and evaluation of the relevant and practical methods applied to the project.

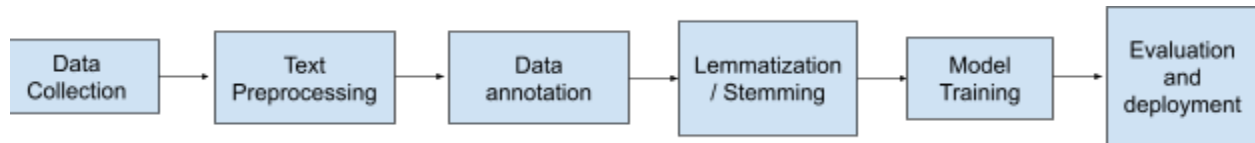


Figure 1 : Methodology step used in this project

3.1 Data Collection

The dataset used for this project is the comment of a tudung product that was scrapped from shopee. The reason that we choose tudung from shopee is because we need to scrape the product that contains both english and malay language, or mix of them. Shopee scraping api is used to search for the top sales tudung product and scrape the 100 tudung shop's product review (Shopee Scraper · Apify, n.d.). As we found that the review from shopee 95% of them is from a 5 star review and only 1 to 2% is from 1 or 2 star review. It will make the dataset scrapped to be seriously unbalanced. From the research from A Comprehensive Guide to 5-Star Surveys, it shows that a score of 5 star means an excellent experience and a highly satisfied customer with the product or service and a score of 2 stars means a poor experience, while a score of 1 star is reserved for extremely negative experiences (Raileanu, 2023). So, we decided to scrape the review separately, which is review from 4 and 5 star and save it in the positive dataset, while save the review scrape from 1 and 2 star rating in the negative dataset. To get the balanced dataset, we scrape 3000 positive and 3000 negative reviews from the shopee tudung products reviews. To get the dataset that is more meaningful, we only scrape the review that contains more than or equal 20 texts. The positive dataset contain about 2,900 reviews, whereas the negative dataset consists of about 2700 reviews. Both datasets contain the Username and Comment as shown in Figure 1.

	Username	Comment
0	faisaparvin82	Material:quality kain Tak berapa elok..\\n\\nKai...
1	syahirahsheikh96	Material:crepe\\nComfortability:susah nak bantu...
2	bbywjy	Kain berbau ..👃 basuh dgn pewangi pun masih ba...
3	a****u	Comfortability:out\\nMaterial:out\\n\\nwhehh teruk...
4	bidadarisweet	material xbest..kain keras & kasar..mmg senng ...

Figure 2 : Sample of dataset

3.2 Data preprocessing:

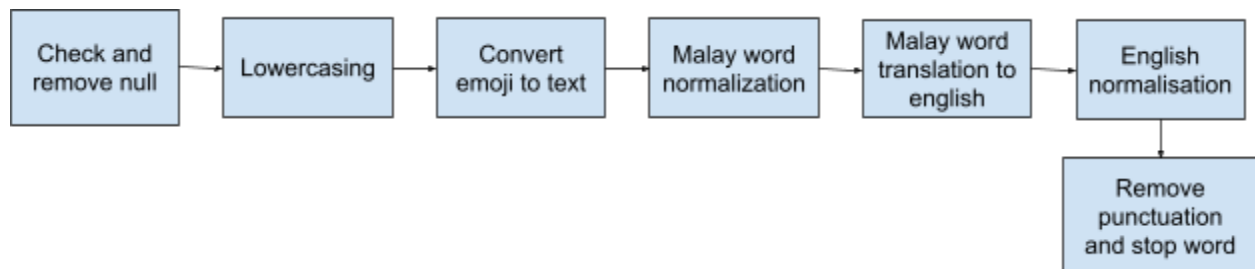


Figure 3 : Preprocessing steps

Preprocessing is important for a sentiment analysis. Based on the research of paper from (The impact of preprocessing steps on the accuracy of machine learning algorithms in sentiment analysis) that use different preprocessing method to find the impact of preprocessing on the accuracy , it shows that the result of accuracy is higher for the dataset that applied the text preprocessing (McSwine, 2023). It is because text preprocessing will increase the quality of the dataset and so machine learning models can learn the pattern and characteristics of language better (Çıtak, 2023). Thus, as mentioned in the previous solution in introduction, text-preprocessing is needed and important for this project. Malaysians especially youth usually like to use short forms and mixed language for texting (INTERNATIONAL YOUNG SCHOLARS JOURNAL OF LANGUAGES, n.d.). Malaysian Rojak language that mixed both English and Malay words in the sentences and the way of Malaysian texting a text make the preprocessing part interesting. Thus, we use various steps to handle the preprocessing of text as shown in figure 3.

3.2.1 Check and remove null

Removing null values from the dataset is one of the important steps in text preprocessing. The null value did not bring any meaning to the dataset. These null values will also affect the performance and accuracy of the machine learning algorithm. So, it is very important to remove null values from the dataset before applying any machine learning algorithm to that dataset. Figure 4 shows that there is no null value in this project, as we only scrape the comment with more than 20 texts.

```
print(positive.isnull().any().any())
print(negative.isnull().any().any())

False
False
```

Figure 4 : no null in the dataset

3.2.2 Lowercase

In the NLP tasks, lowercasing is a crucial step in text preprocessing. Lowercasing converts all the characters to lowercase to ensure the consistency in token representation. This is because the NLP model will treat both words with the same spelling but having different uppercase or lowercase characters as different words. For example, the word 'cantik' and 'Cantik' will be treated as different tokens if lowercasing is not applied. From figure 6, it shows the result after lowercasing is applied in this dataset, 'cantik' and 'Cantik' will be simplified as 'cantik'.

Kain cantik berkualitas..tak jarang..corak cantik..berbaloi beli..cuma tak dimasuk plastik..semata2 harapkan plastik kurier..sayang klu kain rosak dihimpit2 barang-barang kurier dalam kargo..klu ada plastik barang terjamin sikit..anyway tq Seller..

Barang diterima dalam keadaan baik penghantaran pun pantas. Cantik shawl ni mudah je sarung trus siap. Kain pun selesa d pakai, harga pun ok berbaloi la bli. Lain kali boleh repeat order lagi. 🙌🙌🙌🙌🙌

Figure 5 : dataset before lowercasing

kain cantik berkualitas..tak jarang..corak cantik..berbaloi beli..cuma tak dimasuk plastik..semata2 harapkan plastik kurier..sayang klu kain rosak dihimpit2 barang-barang kurier dalam kargo..klu ada plastik barang terjamin sikit..anyway tq seller..

barang diterima dalam keadaan baik penghantaran pun pantas. cantik shawl ni mudah je sarung trus siap. kain pun selesa d pakai, harga pun ok berbaloi la bli. lain kali boleh repeat order lagi. 🙌🙌🙌🙌🙌

Figure 6 : dataset after lowercasing

3.2.3 Emoji to text converting

Emojis and special characters are common in text data, especially in commenting. Processing these elements correctly is crucial for accurate sentiment analysis. Based on research from Front Psychol, using emoji can show emotional or contextual meaning when communicating (A Systematic Review of Emoji: Current Research and Future Perspectives, 2019). So, instead of removing those emoji that can bring more meaning in the comment, we use library emoji to handle this emoji character in the dataset. Thus, the model can interpret the text converted from emoji and make it better in learning and analysis. Emoji in the comment is converted to text as shown in figure 7.

Comment	Emoji_Normalised_Comment
tak tebal dan tak nipis sgt, ironless belum cuba, tapi rse kain sejuk ni mesti selesa tudung dah selamat sampai. dua² cantik ☺ next time boleh repeat order lagi. 🍷	tak tebal dan tak nipis sgt, ironless belum cuba, tapi rse kain sejuk ni mesti selesa tudung dah selamat sampai. dua² cantik :smiling_face: next time boleh repeat order lagi. :thumbs_up_light_skin_tone:
kain cantik berkualiti..tak jarang..corak cantik..berbaloi beli..cuma tak dimasuk plastik..semata2 harapkan plastik kurier..sayang klu kain rosak dihipit2 barang-barang kurier dalam kargo..klu ada plastik barang terjamin sikit..anyway tq seller..	kain cantik berkualiti..tak jarang..corak cantik..berbaloi beli..cuma tak dimasuk plastik..semata2 harapkan plastik kurier..sayang klu kain rosak dihipit2 barang-barang kurier dalam kargo..klu ada plastik barang terjamin sikit..anyway tq seller..
barang diterima dalam keadaan baik penghantaran pun pantas. cantik shawl ni mudah je sarung trus siap. kain pun selesa d pakai, harga pun ok berbaloi la bli. lain kali boleh repeat order lagi. 🍷🍷🍷🍷🍷	barang diterima dalam keadaan baik penghantaran pun pantas. cantik shawl ni mudah je sarung trus siap. kain pun selesa d pakai, harga pun ok berbaloi la bli. lain kali boleh repeat order lagi. :thumbs_up::thumbs_up::thumbs_up::thumbs_up::thumbs_up:
tq.. penghantaran cepat... sebab dekat kan... tapi shoope express je yang lambat... makin tak ok shoope express kat sbnnnnnnnn ni sekarang 😞 aritu jnt skrg shope lepas tu mungkin ape pun xthlaa 🍷 tapii bestla tudung ni.sesuai dgn harga.. boleh pakai g jmnt ank2 blk sek.. cantik2.. nnti beli lg ye..	tq.. penghantaran cepat... sebab dekat kan... tapi shoope express je yang lambat... makin tak ok shoope express kat sbnnnnnnnn ni sekarang:pensive_face: aritu jnt skrg shope lepas tu mungkin ape pun xthlaa :unamused_face: tapii bestla tudung ni.sesuai dgn harga.. boleh pakai g jmnt ank2 blk sek.. cantik2.. nnti beli lg ye..
maaf gambar tiada kaitan. 🙏 barang diterima dalam keadaan bagus. semua colour yg di order ada. tudung sangat cantik dan senang pakai. harga pun berpatutan, mampu milik. lain kali akan order lagi. terima kasih seller. semoga sentiasa di. urahkan rezeki. insyallah. aamiin.	maaf gambar tiada kaitan. :folded_hands: barang diterima dalam keadaan bagus. semua colour yg di order ada. tudung sangat cantik dan senang pakai. harga pun berpatutan, mampu milik. lain kali akan order lagi. terima kasih seller. semoga sentiasa di. urahkan rezeki. insyallah. aamiin.

Figure 7 : result after convert emoji

3.2.4 Malay word normalization

Before translating the malay word to english, malay word normalization is needed to handle the malay word short form as the translation library did not understand the various styles and variations in abbreviations, slang, or short forms that may bring the same meaning but appear differently. In this project, we use the malay abbreviation dataset to convert the various styles of word in abbreviations, slang, or short forms in malay word to a standard malay word. After converting it, the dictionary can correctly translate the standard Malay word into English and the sentiment analysis models can correctly interpret and understand the deep meanings of the word. For example, the word ‘trm ksh’ and ‘terima kasih’ will convert to ‘terima kasih’ and will be translated to thank you in the following translation step. The result after malay normalization is shown in figure 8.

Emoji_Normalised_Comment	Malay_Normalised_Comment
tak tebal dan tak nipis sgt ironless\nbelum cu...	tak tebal dan tak nipis sangat ironless belum ...
kain cantik berkualititak jarangcorak cantikbe...	kain cantikkah? berkualititak jarangcorak cant...
barang diterima dalam keadaan baik penghantara...	barang diterima dalam keadaan baik penghantara...
👉 penghantaran cepat sebab dekat kan tapi sho...	terima kasih penghantaran cepat sebab dekat ha...

Figure 8 : result after malay normalize

3.2.5 Translation malay to english

Based on the research in technique used in the research background it shows that there is no difference for training two language models and training models using only one language after translation. So, we will use translation methods and train the model for only using one language. The translation process is crucial for our project preprocessing as our dataset contains both Malay and English words. For the model to train for sentiment analysis, we translate the Malay word to English and train using the english language only. So, by using the translation of malay words to english, it simplified the learning process of machine language as it only needed to understand and analyze the text in english. In addition, it also makes the data annotation and other methods easier to process as many of the libraries, tools and api are optimized for english text. We use the 'mtranslate' library, which is the library in python used to translate text using google's translation. The translated result is shown in figure 9.

Malay_Normalised_Comment	Translated_Comment
quality kain tak berapa elok kain licin	the fabric quality is not very good, the fabri...
crepe susah hendak bentuk takpadan dengan harg...	Crepe is hard to find the shape that doesn't m...
kain berbau grinningfacewithsweat basuh dengan...	cloth that smells like grinningfacewithsweat e...
buat latihan buat latihan wehh teruk sangat qu...	do practice do practice wehh very bad qualityl...
material xbestkain keras kasarmmg senng bntukl...	The material is made of hard, hard cloth, so i...

Figure 9 : Result after malay word translated to english

3.2.6 English normalization

AS the dataset contains both english and malay language or the mix of both, so we need to handle the english normalization to convert the informal style of english word to the standard word. This step is similar to the Malay normalization but it is applied to the English word. After this English normalization, it makes the model more efficient in training as the words that have the same meaning but appear different, will be converted to the same. In this project, the library contractions are used in English word normalization. The result of English normalization is shown in figure 10.

Translated_Comment	Normalized_Comment
not thick and not very thin, ironless, haven't...	not thick and not very thin, ironless, have no...
Is the fabric beautiful? rare quality, beautif...	Is the fabric beautiful? rare quality, beautif...
The item was received in good condition and th...	The item was received in good condition and th...
thank you for the fast delivery because it's c...	thank you for the fast delivery because it is ...

Figure 10 : result of text normalization

After that, in this text normalization process, we do the spell check and remove meaningless text as we found that there is some typo error in the dataset. After spell correction, we check the meaningless word and remove it , so that the text that is not able to be preprocessed into standard word will be removed. This can prevent the not preprocessed word to be learned or analyzed by the machine model. The result after spelling correct and meaningless text removal is shown in figure 11.

Normalized_Comment	Normalized2_Comment	Meaningful_Comment
not thick and not very thin, ironless, have no...	not thick and not very think ironless, have no...	not thick not very think have not tried it yet...
Is the fabric beautiful? rare quality, beautif...	Is the fabric beautiful rare quality beautiful...	Is fabric beautiful rare quality beautiful pat...
The item was received in good condition and th...	The item was received in good condition and th...	item was received in good condition delivery w...
thank you for fast delivery because it is clos...	thank you for fast delivery because it is clos...	thank fast delivery it is close but express is...

Figure 11 : result after correct spelling and remove meaningless word

3.2.7 Data cleaning

In the data cleaning step, we remove the punctuation and stopwords. In punctuation removal, the punctuation such as ,#/()&^ that do not bring meaning and are not important in machine learning will be removed to reduce noise. For stopwords removal, the stop word like a, the, and , or , of will be removed as they add little value in determining the sentiment and will negatively affect accuracy. This is because stop words like the, or, and will always be used in the sentences. So this stopword will make the machine model to learn the unnecessary words and affect the performance and accuracy of the model. But there are some stop words like not,no,never is very important for sentiment analysis as this negation word will change the meaning of the sentences. For example, if the negation is removed, 'not good' will be converted to 'good', which is the opposite meaning for the original word. So, in stopword removal, we do not remove the negation word. The result after data cleaning is shown in figure 12

	Cleaned_Comment
0	not thick not think not tried yet think cool f...
1	fabric beautiful rare quality beautiful patter...
2	item received good condition delivery fast bea...

Figure 12 : result after data cleaning

3.3 Data annotation (VADER)

Data annotation is the process of labeling the sentiment of the comment by using pre-trained machine learning or manually labeling by human. Data annotation is important in machine learning as it provides the labeled example for the machine learning model to learn the pattern and relationship between text. Based on the research that stated in the research background, we found that automatic annotation is better for our project as it increases the quality and efficiency of the project. In addition, we found that the label “neutral” is considered as a lack of opinion when doing research on research background.

So, in our project, we decided to use the automatic annotation to label the preprocessed comment as positive and negative. For automatic annotation, we use the pretrained annotation library to label the comment. As there are various existing libraries of data annotation, we use different pretrained tools to compare which one is the better and find the best tools. The tools that we used are textBlob, vader, and affin. Textblob and affin will be done by others teammates, the following will focus on the vader library.

VADER (Valence Aware Dictionary and Sentiment Reasoner) is a lexicon and rule-based sentiment analysis tool that is specifically attuned to sentiments expressed in social media. It is fully open-sourced (Hutto, n.d.). VADER is able to detect the polarity of sentiment for a given text and analyze the sentiment (positive or negative) on it. VADER uses a lexicon of sentiment-related words to determine the overall sentiment of a text. In figure 13 show the example of how the lexicon is structured, with each word having a valence rating. This labeled lexicon is built using Amazon's Mechanical Turk, which is a platform that pays ‘crowdworkers’ to perform tasks together and resulting in an impressively efficient method for doing this.

Word	Sentiment rating
Tragedy	-3.4
Rejoiced	2
Insane	-1.7
Disaster	-3.1
Great	3.1

Figure 13 : VADER example

VADER function is by assigning polarity scores to the individual words or phrases, generating a compound score for the text's overall sentiment. The compound score that ranges from -1 to +1, encapsulates the text's aggregate sentiment. From the VaderSentiment scoring part, VADER is used where a compound score higher than 0.5 is labeled as a positive sentiment, while a score lower or equal to 0.5 is interpreted as negative (Vader Sentiment, 2019). Figure 14 shown the result after apply the VADER

	Cleaned_Comment	Score	negative	neutral	positive	compound
0	not thick not think not tried yet think cool f...	{'neg': 0.071, 'neu': 0.573, 'pos': 0.356, 'co...	0.071	0.573	0.356	0.8726
1	fabric beautiful rare quality beautiful patter...	{'neg': 0.17, 'neu': 0.505, 'pos': 0.324, 'com...	0.170	0.505	0.324	0.7783
2	item received good condition delivery fast bea...	{'neg': 0.0, 'neu': 0.505, 'pos': 0.495, 'comp...	0.000	0.505	0.495	0.9559
3	thank fast delivery close express thing slow n...	{'neg': 0.042, 'neu': 0.485, 'pos': 0.474, 'co...	0.042	0.485	0.474	0.9618
4	sorry picture nothing folded hands item receiv...	{'neg': 0.031, 'neu': 0.541, 'pos': 0.428, 'co...	0.031	0.541	0.428	0.9509

Figure 14 : Results after applying VADER

After sentiment annotation by VADER, we identify the most frequently occurring words from both of the positive and negative comments. Then, we relabel the labeled comment using these frequently occurring words. If a comment consists of more negative words than positive words, it will relabel the comment as 'negative', and if the comment consists of higher frequency of positive words, it will relabel the comment as 'positive'. This is because VADER has some limitations that do not perform well in complex sentences. It relies heavily on the lexicon, so it only cares about individual words and completely ignores the context in which it is used.

For example, “the party was savage” will be negative when considered by any token-based algorithms (Sentiment Analysis in Python: TextBlob Vs Vader Sentiment Vs Flair Vs Building It From Scratch, n.d.). So, we use this additional step to do the data annotation to get a more accurate annotation result. The result of final data annotation using VADER is shown in figure 15.

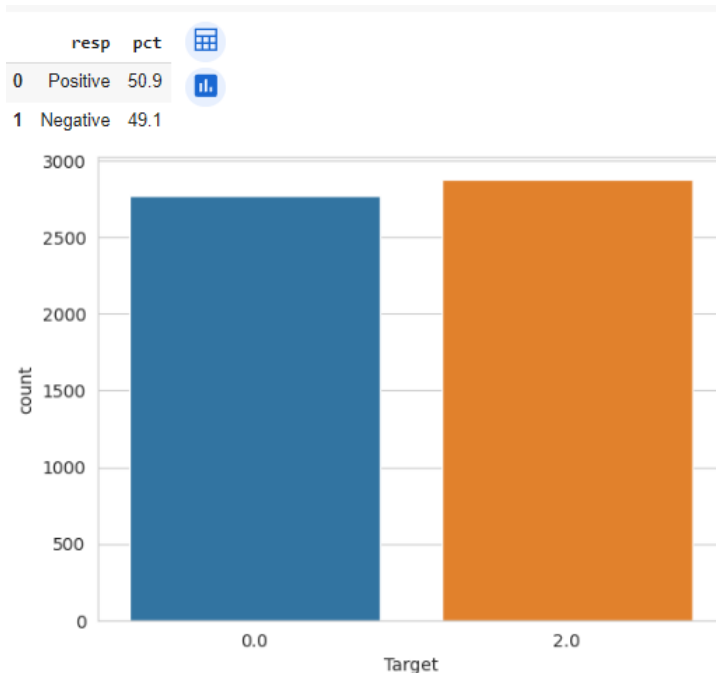


Figure 15 : result of final VADER dataset

3.4 stemming/lemmatization

Stemming is a straightforward normalization technique, most often implemented as a series of rules that are progressively applied to a word to produce a normalized form. It removes the prefix and suffix of the word, and the word after stemming will be not a valid word. For example, the word ‘troubles’ will stem into ‘troubl’ which is not a valid English word. For lemmatization, it is similar to the stemming but a more complex version. It reduces each word to its proper base form, which is a valid word that we can find in a dictionary. Stemming or lemmatization is a crucial step in the machine learning preprocessing process to simplify the word into their root or base word. This is important as it simplifies the representation of text data and can lead to more efficient and effective models. Based on the research in the research background, we found that both stemming and lemmatization has their own limitations. So, instead of using one of the tools, we use both of them and 1 mix of them. 3 sets of data will be created for training by using first stemming technique, second lemmatization technique, and third mix technique. Before performing this lemmatization and stemming, tokenization is handled. Tokenization breaks down the text into individual words or tokens and it is crucial for feature extraction. Tokenization divides the text into smaller units, making it easier for further analysis. The result of stemming is shown in figure 16, lemmatization in figure 17 and mix of lemmatization and stemming in figure 18. Each of these dataset is then implemented into the training model and figure out which technique has the best accuracy.



Figure 16 : Stemming wordnet

[illegible][illegible]

18

3.5 feature engineering

Feature extraction is an important step in a NLP project, it transforms the textual data into a mathematical representation, which is in the form of vectors known as word embeddings. It helps algorithms to use the importance of the words to predict outcomes. A survey showed that most of the text-based recommender systems in the libraries use TF-IDF for extracting textual features (Mitra, n.d.). In addition, papers that compare N-Gram with TF-IDF (A Comparative Study on TF-IDF Feature Weighting Method and Its Analysis Using Unstructured Dataset, n.d.) and papers that compare TF-IDF and Hashing Vectorizer (Roshan, 2023) showed that the TF-IDF feature provides more performance than other features and it has the maximum accuracy among them. So, we will use TF-IDF for our feature extraction to generate numeric form for our data. TF-IDF works by counting the frequency of the word and if the word that is often used in the sentences like 'material' and 'product' in our project, it will give a lower rank. The higher the rank, the more important the word is in the comment. It is counted by multiplying the TF and IDF.

$$\text{tfidf}(w, d, D) = \text{tf}(w, d) \times \text{idf}(w, D)$$

TF (frequency term) is the frequency of the word occurrence in the whole dataset. It is the ratio of a number of words per the total number of words in the whole dataset. For example, if the sentence has 6 words and contains two "the", the TF ratio of this word would be (2/6).

$$\text{tf}(w, d) = \frac{\text{occurrence of } w \text{ in document } d}{\text{total number of words in document } d}$$

IDF (Inverse Document Frequency) is used to calculate the importance of the word based on the whole dataset. It is calculated by dividing the total number of comments in the dataset by the number of sentences containing the word.

$$\text{idf}(w, D) = \ln\left(\frac{\text{total number of documents } (N) \text{ in corpus } D}{\text{number of documents containing } w}\right)$$

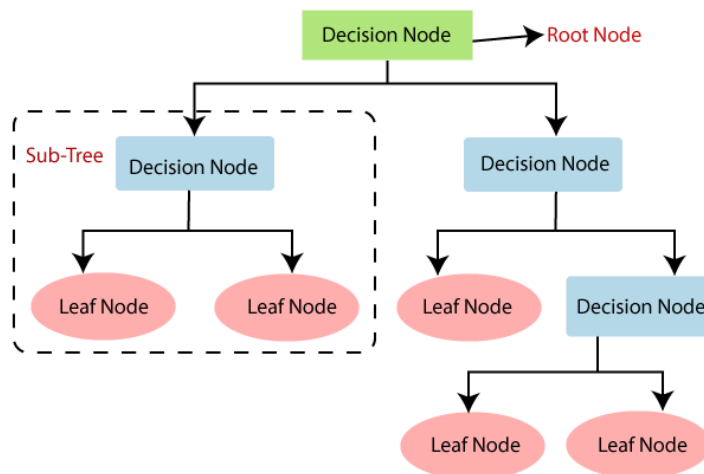
$$\text{idf}(w, D) = \log\left(\frac{N}{f(w, D)}\right)$$

3.6 Model Selection

In the machine learning model in this project, we use various machine learning models to train the 3 dataset that have been preprocessed before. The model included :

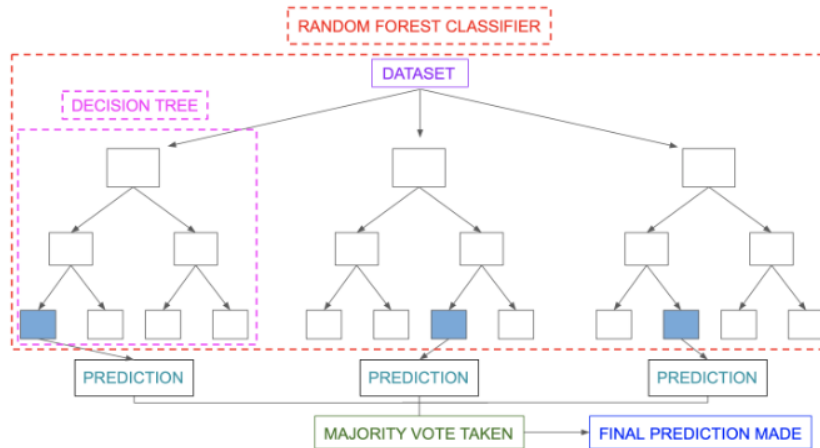
1. Decision Tree

A decision tree is a supervised learning algorithm for classification and regression tasks. It has a hierarchical tree structure that consists of root nodes, branches, internal nodes, and leaf nodes where internal nodes represent the features of a dataset, branches represent the decision rules and each leaf node represents the outcome.



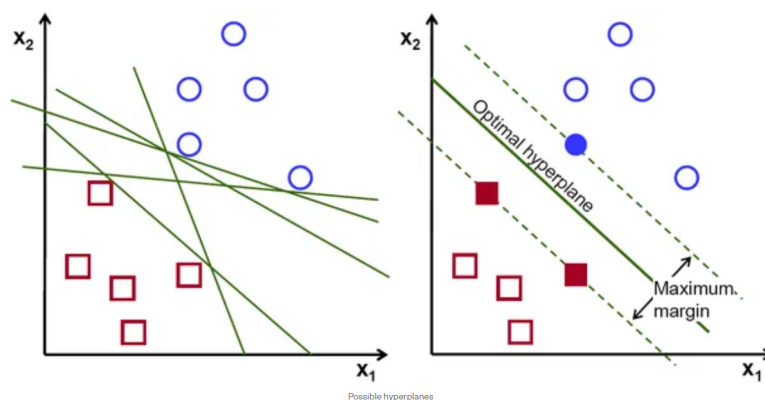
2. Random Forest

Random Forest is one of the supervised learning algorithms. In a random forest classification, multiple decision trees are created using different random subsets of the data and features. Each decision tree is like an expert, providing its opinion on how to classify the data. Predictions are made by calculating the prediction for each decision tree. It predicts by taking the average or mean of the output from various trees. Increasing the number of trees increases the precision of the outcome.



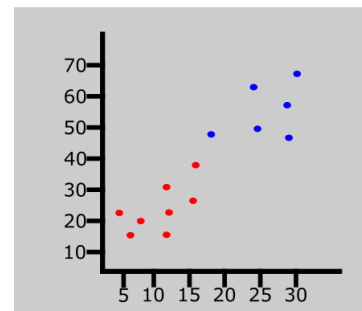
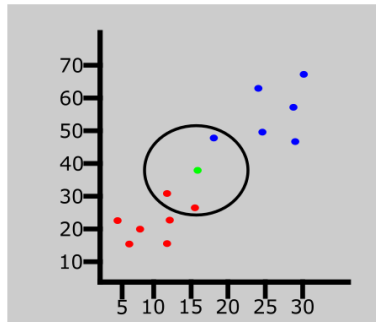
3. Support Vector Machine

Support Vector Machine (SVM) is machine learning to find a hyperplane in an N-dimensional space (N — the number of features) that distinctly classifies the data points. To separate the two classes of data points, there are many possible hyperplanes that could be chosen. Our objective is to find a plane that has the maximum margin, or the maximum distance between data points of both classes. Maximizing the margin distance provides some reinforcement so that future data points can be classified with more confidence. Hyperplanes are decision boundaries that help classify the data. Data points falling on different sides of the hyperplane are treated as different classes.



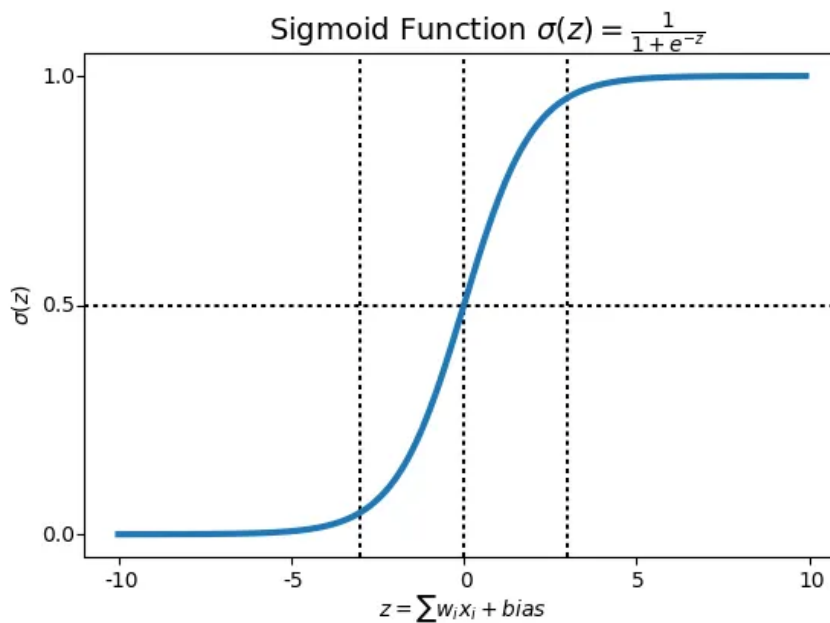
4. K-Nearest Neighbors

K-Nearest Neighbour Classifier (KNN) is a supervised learning algorithm used in regression or classification. It is often used in statistical estimation and pattern recognition. The KNN algorithm is similar to a voting system, where the majority class label determines the class label of a new data point among its nearest 'k' (where k is an integer) neighbors in the feature space. First, the k value which is the neighbors the algorithm takes into account while producing a prediction is selected. KNN relies on measuring the distance between data points in a feature space such as Euclidean distance or Manhattan distance. For a new data point, KNN calculates the distance between that point and all the points in the training dataset. The next step is to choose the K data points that are closest to the input point. Then, it assigns the class label that occurs most frequently among the K neighbors. KNN assigns this as the prediction for the input data point.



5. Logistic Regression

Logistic Regression is one of the most common statistical methods in supervised learning classification algorithms, used to analyze the dataset with one or more independent variables and determine an outcome. It predicts binary classification problems, where the outcome is a categorical variable with two classes, typically labeled as 0 or 1, Yes or No, positive or negative and so on. The logistic regression model uses the logistic function (also known as the sigmoid function) to model the probability that the input belongs to one of the two class. It will produce an S-shaped curve whose range is between 0 and 1. The logistic regression model will calculate the odds (Probability of Event / Probability of Not Event) of the event which belongs to class 1 and transform it into the probability score. If the probability is greater than the chosen threshold (default 0.5), the model predicts class 1, else it will predict class 0.



Sigmoid Function Graph

$$f(x) = \frac{1}{1 + e^{-(x)}}$$

Formula of a sigmoid function | Image: Analytics India Magazine

6. Multinomial Naive Bayes

Multinomial Naive Bayes algorithm is a probabilistic learning that is mostly used in Natural Language Processing (NLP). The algorithm is based on the Bayes theorem and predicts the tag of a text. It calculates the probability of each tag and then gives the tag with the highest probability as output. Bayes theorem, formulated by Thomas Bayes, calculates the probability of an event occurring based on the prior knowledge of conditions related to an event.

$$P(A|B) = P(A) * P(B|A)/P(B)$$

$P(B)$ = prior probability of B

$P(A)$ = prior probability of class A

$P(B|A)$ = occurrence of predictor B given class A probability

In modeling ,the dataset is split to the training set and testing set . The training set is used to feed and train the system or machine so that it can learn the pattern of the data and the machine or system should produce an accurate result based on the training set pattern. While the test set will be used to evaluate the ability of a system to deal with different kinds of data and produce valid and reliable results. Based on the accuracy table in figure 19, the 80:20 data splitting ratio produces the highest test set accuracy, making it the better data splitting choice. So we choose the 80:20 splitting choice and make the training after splitting it.

8:2 dataset				7:3 dataset				6:4 dataset			
	Model	Accuracy	Time		Model	Accuracy	Time		Model	Accuracy	Time
0	DT	0.837456	0.450927		DT	0.830878	0.390617		DT	0.842687	0.206747
1	RF	0.893993	4.747698		RF	0.885681	5.354632		RF	0.891737	3.047005
2	SVM	0.898410	2.489796		SVM	0.887448	2.705682		SVM	0.885108	2.033608
3	KNN	0.826855	1.205169		KNN	0.829110	2.564280		KNN	0.833849	2.374886
4	LRM	0.913428	0.069317		LRM	0.905716	0.105959		LRM	0.902784	0.076725
5	MNB	0.863074	0.005085		MNB	0.856217	0.004168		MNB	0.855502	0.008689

Figure 19 : accuracy of different splitting choice.

```
models = {}
models['DT'] = DecisionTreeClassifier(max_depth=None)
models['RF'] = RandomForestClassifier(max_depth=50, n_estimators=250)
models['SVM'] = SVC(kernel='poly', gamma='scale', C=1, degree=2)
models['KNN'] = KNeighborsClassifier(n_neighbors=10)
models['LRM'] = LogisticRegression(multi_class='auto', solver='saga', C=10)
models['MNB'] = MultinomialNB(alpha=1.0)
```

Figure 31 : hyper parameter used for each model

Hyperparameter tuning is critical to achieving optimal performance in machine learning models. A poorly tuned model can result in underfitting or overfitting, which can lead to inaccurate predictions and poor performance on new data. The best parameter of each model is searched using GridSearchCV and set a predefined range of values. Figure 31 shows all the best parameters for each machine learning model.

3.7 Evaluation and deployment

Four measures are utilized to evaluate the performance of the classifiers: accuracy, precision, recall, and F1-score is used.

Accuracy score in machine learning is an evaluation metric that measures the number of correct predictions made by a model in relation to the total number of predictions made. (Bressler, 2023)The formula of accuracy is

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

Precision is the ratio $tp / (tp + fp)$ where tp = number of true positives and fp = number of false positives. The precision is the ability of the classifier not to label as positive a sample that is negative.(Sklearn.Metrics.Precision_Score, n.d.).The formula of accuracy is

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall is The recall is the ratio $tp / (tp + fn)$ where tp is the number of true positives and fn the number of false negatives. The recall is the ability of the classifier to find all the positive samples.(Sklearn.Metrics.Recall_Score, n.d.). The Recall Score formula is

$$\text{Recall} = \frac{TP}{TP + FN}$$

The F1 score can be interpreted as a harmonic mean of precision and recall, where an F1 score reaches its best value at 1 and worst score at 0. The relative contribution of precision and recall to the F1 score are equal (Sklearn.Metrics.F1_Score, n.d.). The F1 Score formula is

$$F1\ Score = 2 \times \frac{recall \times precision}{recall + precision}$$

For deployment, streamlit is used after selecting the best model after model evaluation. Streamlit is an open-source app framework in python language that helps to create beautiful web apps for data science and machine learning in a little time.

4.0 Results

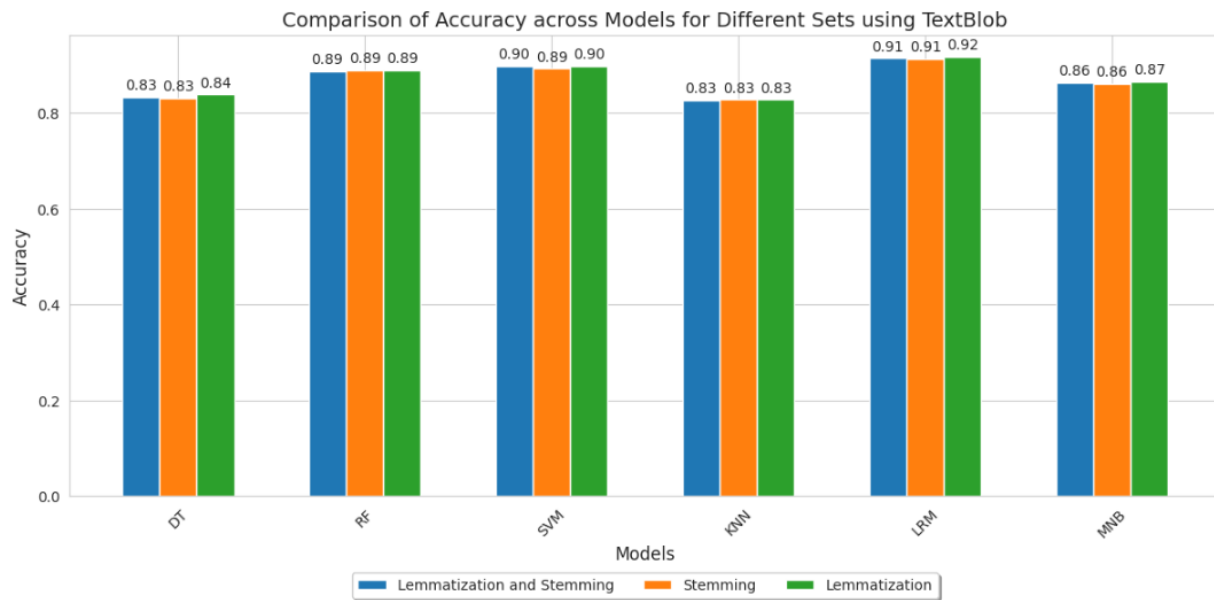


Figure 20 : Graph of textblob dataset accuracy

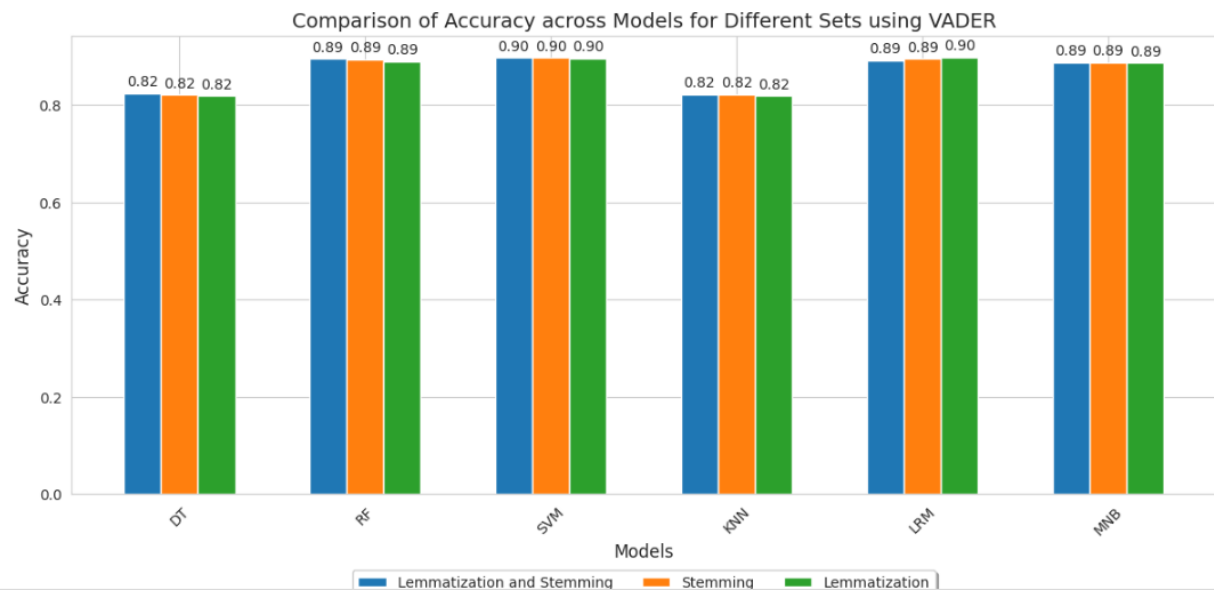


Figure 21 : Graph of VADER dataset accuracy

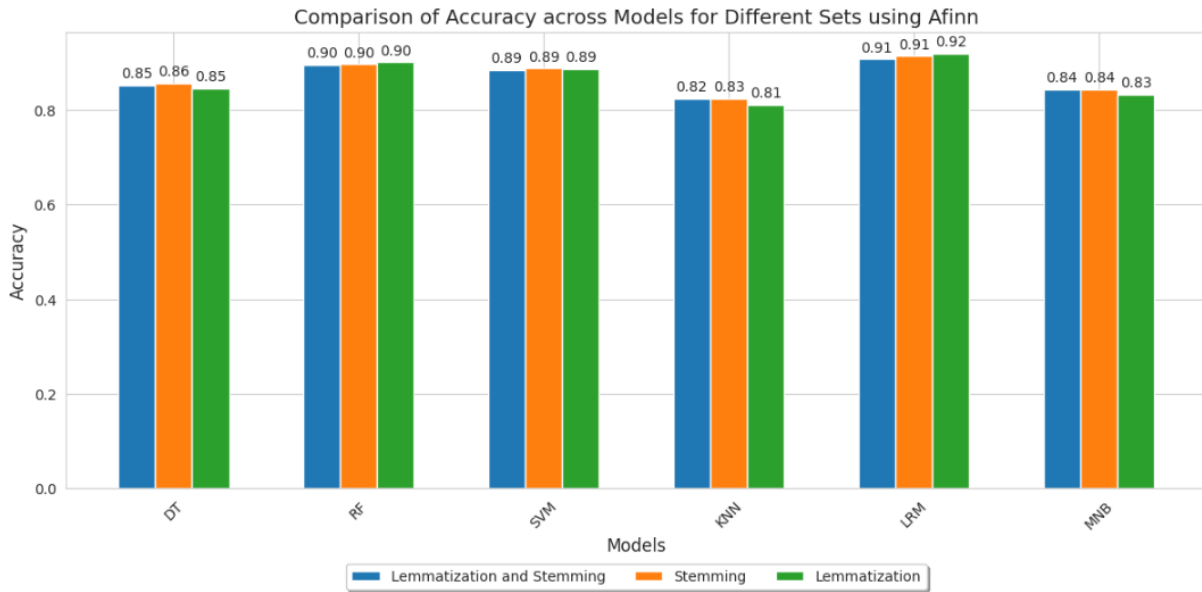


Figure 22: Graph of Affin dataset accuracy

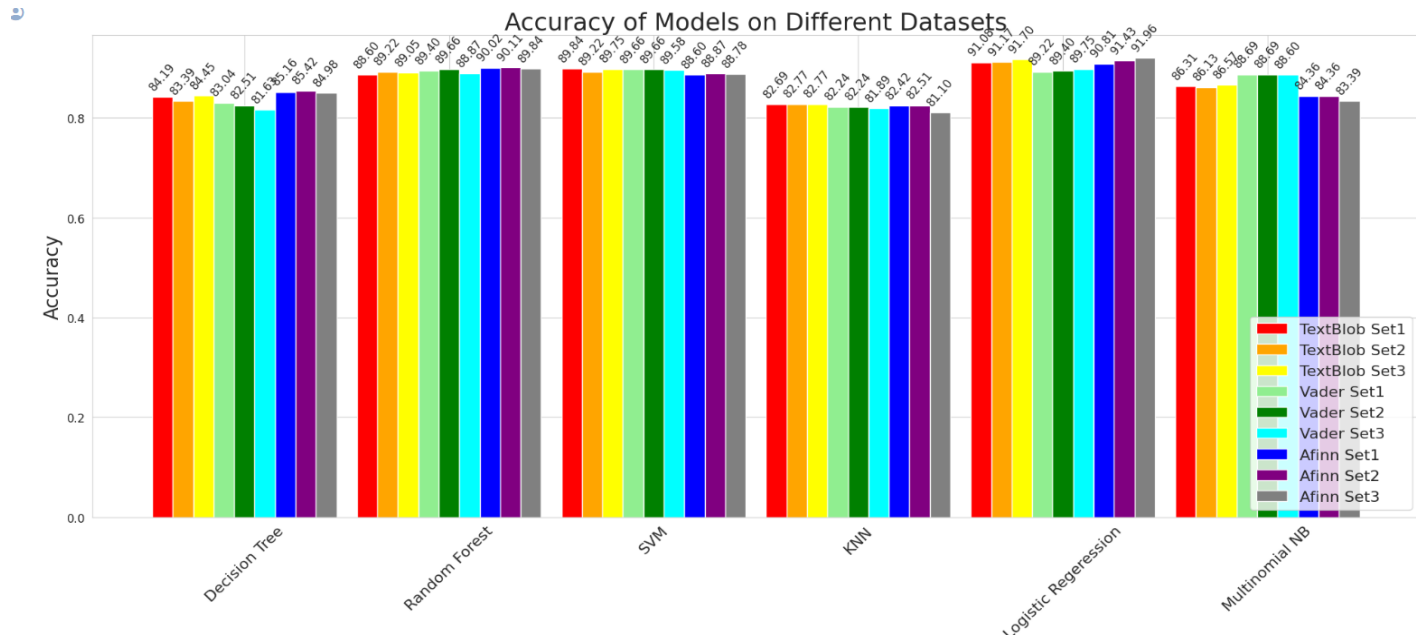


Figure 23 : Graph of accuracy for all data annotation

	textblob			vader			affin		
	mixed	stem	lemmatize	mixed	stem	lemmatize	mixed	stem	lemmatize
DT	84.19%	84.45%	83.39%	83.04%	82.51%	81.63%	85.16%	85.42%	84.94%
RF	88.60%	89.22%	89.05%	89.40%	89.66%	88.87%	90.02%	90.11%	89.84%
SVM	89.84%	89.22%	89.75%	89.66%	89.66%	89.58%	88.60%	88.87%	88.87%
KNN	82.69%	82.77%	82.77%	82.24%	82.24%	81.89%	82.51%	81.10%	82.42%
LRM	91.08%	91.17%	91.70%	89.22%	89.40%	89.75%	90.81%	91.43%	91.96%
MNB	86.31%	86.13%	86.57%	88.69%	88.69%	88.60%	84.36%	83.39%	84.36%

Table 1 : accuracy of learning model that using different data annotation

Based on the result shown in table 1, all the accuracy of all models is above 80% which means that all the models are well-trained with the dataset. In addition, LRM trained with a dataset that is preprocessed using affin and lemmatization has the highest result on accuracy 91.86%. On the other hand, KNN model which uses stemming and affin in preprocessing has the lowest accuracy score which is 81.10%. The higher the accuracy, the better the model at making correct predictions on the given dataset.

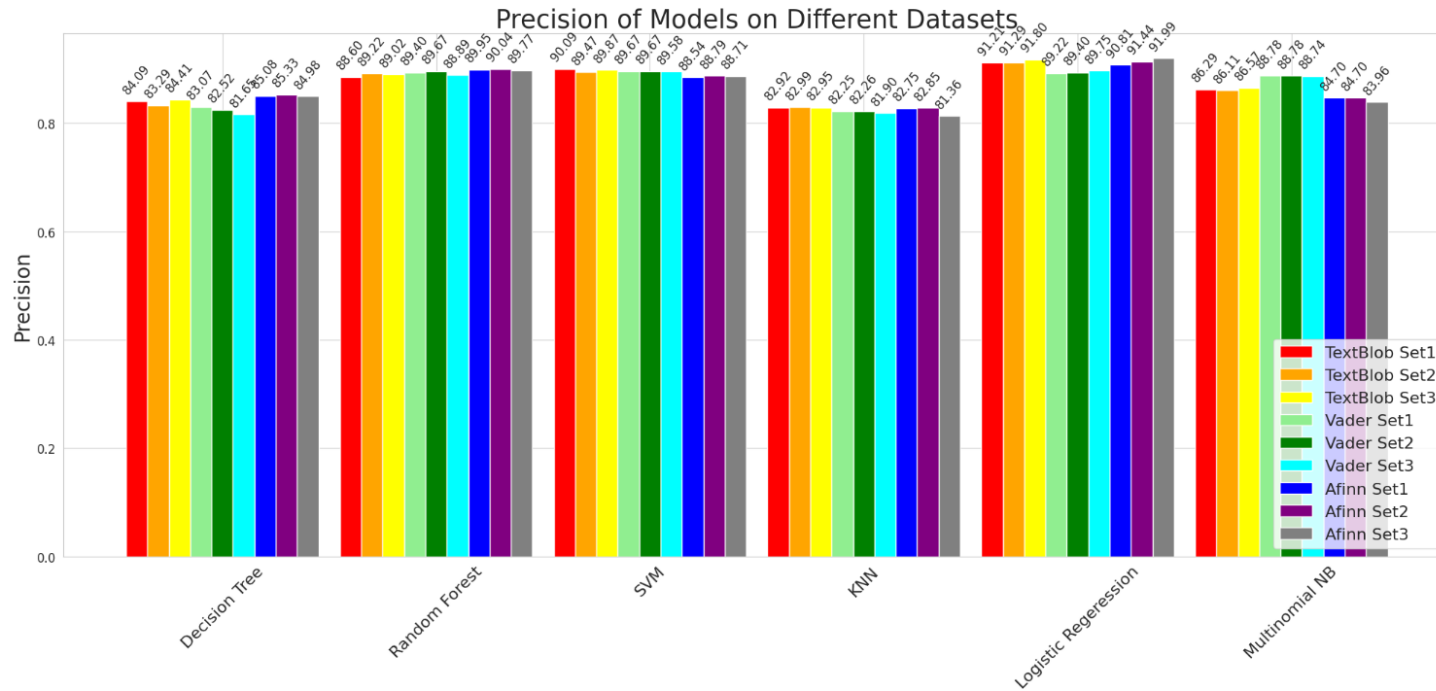


Figure 24 : Graph of precision for all data annotation

	textblob			vader			affin		
	mixed	stem	lemmatize	mixed	stem	lemmatize	mixed	stem	lemmatize
DT	84.09	83.29	84.41	83.07	82.52	81.65	85.08	85.33	84.98
RF	88.6	89.22	89.02	89.40	89.67	88.89	89.95	90.04	89.77
SVM	90.09	89.47	89.87	89.67	89.67	89.58	88.54	88.79	88.71
KNN	82.92	82.99	82.95	82.25	82.26	81.90	82.75	82.85	81.36
LRM	91.21	91.29	91.80	89.22	89.40	89.75	90.81	91.44	91.99
MNB	86.29	86.11	86.57	88.78	88.78	88.74	84.70	84.70	83.69

Table 2 : precision of learning model that using different data annotation

From table 2, it can be shown that the precision of all the models used are 80% or above. This indicates that all the models are quite precise and consistent. Logistic Regression that is preprocessed using affin and lemmatization has the highest precision of 91.99%, while KNN model that is preprocessed using affin and lemmatization has the lowest precision of 81.36%. The higher the precision score, the closer the detected results are going to be.

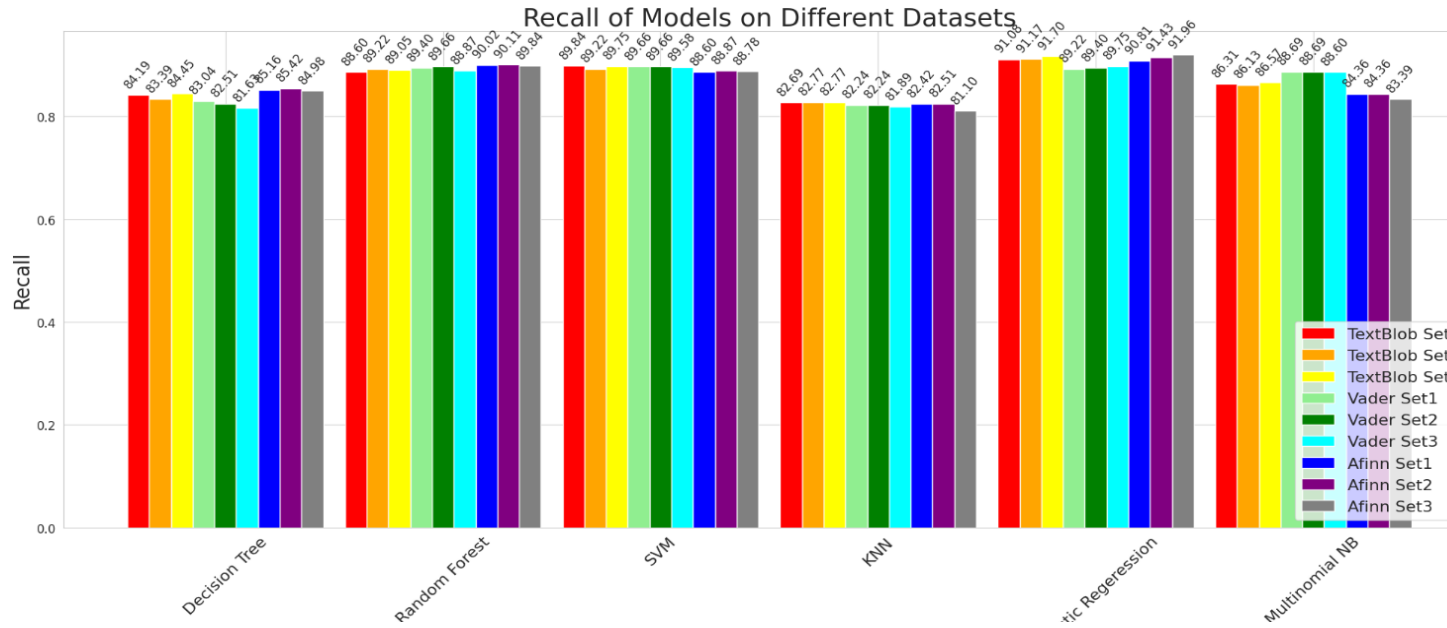


Figure 25 : Graph of recall for all data annotation

	textblob			vader			affin		
	mixed	stem	lemmatize	mixed	stem	lemmatize	mixed	stem	lemmatize
DT	84.19	83.39	84.45	83.04	82.51	81.63	85.16	85.42	84.98
RF	88.60	89.22	89.05	89.40	89.66	88.87	90.02	90.11	89.84
SVM	89.84	89.22	89.75	89.66	89.66	89.58	88.60	88.87	88.78
KNN	82.69	82.77	82.77	82.24	82.24	81.89	82.42	82.51	81.10
LRM	91.08	91.17	91.70	89.22	89.40	89.75	91.81	91.43	91.96
MNB	86.31	86.13	86.57	88.69	88.69	88.60	84.36	84.36	83.39

Table 3 : recall of learning model that using different data annotation

From table 3, it can be seen that the recall of all the models used are more than 80%. Logistic Regression that is preprocessed using affin and lemmatization shows the highest recall score among others which is up to 91.96%. On the other hand, the model that has the lowest recall is Decision Tree model that using vader and lemmatization which is only 81.63%. The higher the recall, the more the positive class is being captured.

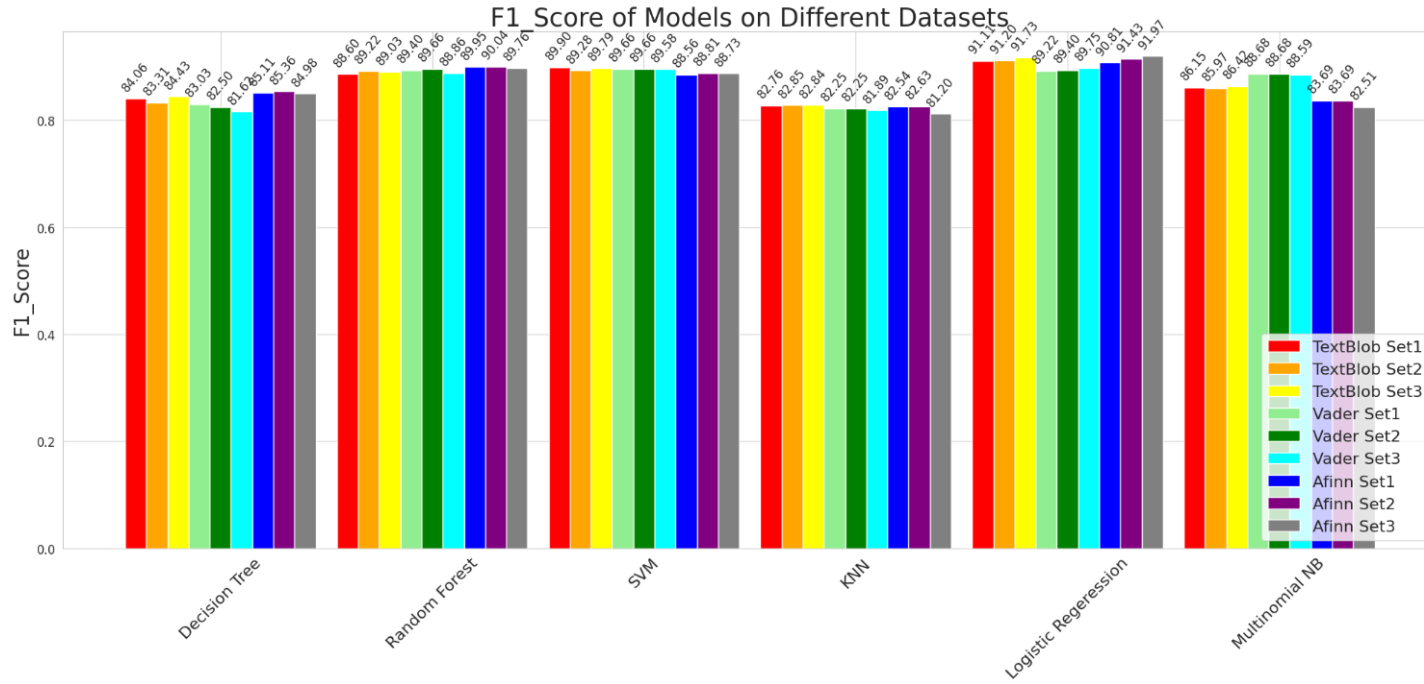


Figure 26 : Graph of f1-score for all data annotation

	textblob			vader			affin		
	mixed	stem	lemmatize	mixed	stem	lemmatize	mixed	stem	lemmatize
DT	84.06	83.31	884.43	83.03	82.50	81.62	82.11	85.36	84.98
RF	88.60	89.22	89.03	89.40	89.66	88.86	89.95	90.04	89.76
SVM	89.90	89.28	89.79	89.66	89.66	89.58	88.56	88.81	88.73
KNN	82.76	82.85	82.84	82.25	82.25	81.89	82.54	82.63	81.20
LRM	91.11	91.20	91.73	89.22	89.40	89.75	90.81	91.43	91.97
MNB	86.15	85.97	86.42	88368	88.68	88.59	83.69	83.69	82.51

Table 4 : f1-score of learning model that using different data annotation

The table 4 shows that the F1-score of all the models used are more than 80%. The F1-score of Logistic Regression that is preprocessed using affin and lemmatization is 91.97%, which is the highest one compared to other algorithms. While the Decision Tree model that used lemmatization and Vader has the lowest F1-score of 81.62%. When the F1 score is high, it indicates that the model's performance in terms of precision and recall is good.

5.0 Discussion and Conclusion

After data evaluation, we found that the best data annotation technique is using affin tools as the highest accuracy, precision, recall and f1-score appear in the model that the data annotation tools used is affin in the preprocessing process. But in the further studies, we found that textblob is more suitable for our dataset in this project. This is because in figure 26, we can see that the vader is biased to negative as it predicts more negative than the other 2 tools in the positive dataset. In addition, figure 27 showed that the affin is more biased in labeling the positives it predicts more positives resulting in a negative dataset.

resp pct			resp pct			resp pct		
0	Positive	91.8	0	Positive	82.5	0	Positive	90.3
1	Negative	8.2	1	Negative	17.5	1	Negative	9.7
Textblob			vader			affin		

Figure 26 : result of positive dataset after data annotation

resp pct			resp pct			resp pct		
0	Positive	41.3	0	Positive	21.4	0	Positive	43.2
1	Negative	58.7	1	Negative	78.6	1	Negative	56.8
Textblob			vader			Affin		

Figure 27 : result of negative dataset after data annotation

In comparing lemmatization,stemming, and mixing of them, we found that lemmatization is the best tool to use. This can be shown when evaluation in result, the lemmatization technique used in preprocessing trained the model that has the highest accuracy, precision, recall and f1-score.

From the evaluation, we also see that logistic regression is the best model as it has the highest accuracy, precision, recall and f1-score. It is expert in both minimizing false positives and false negatives, making it the best algorithm for predicting sentiment.

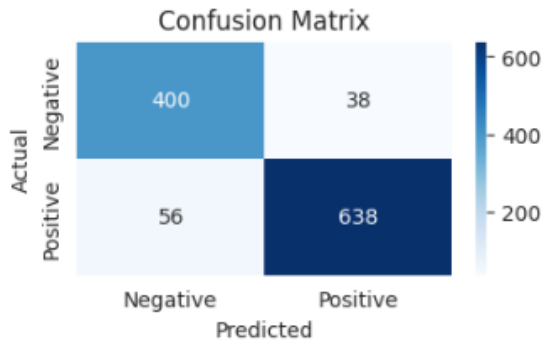


Figure 28 : confusion matrix details for the highest accuracy model, Logistic Regression

In summary, in data annotation, Textblob is chosen. In word simplify technique, lemmatization is used. In machine model learning, Logistic regression is used. In the confusion matrix shown in the figure 24, it shows that the true positive (correctly predicted positive) is 638, while the false positive (incorrectly predicted positive) is 38. This means that the model correctly identified 638 positive classes and the model predicted 38 as positive class, but the actual class was negative. On the other hand, the false negative is 56 and the true negative is 400. This means that the model correctly identified 400 negative classes but predicted 56 positive classes as negative. From the confusion matrix, it can indicate that the Logistic Regression model makes more correct predictions than wrong predictions when testing.

When doing deployment, we found that the model will bias to positive, so after that we found out that the problem is because of the biased dataset of positive class. So we apply the oversampling technique for the improvement of our sentiment analysis system. RandomOverSampler is used and it will randomly duplicate the example of the minority class which is negative class. The result of after oversampling in the highest accuracy model is shown in figure 29.

	Model	Accuracy	Time
0	DT	0.896900	0.490948
1	RF	0.914924	6.273162
2	SVM	0.909877	3.336383
3	KNN	0.839221	1.806085
4	LRM	0.912040	0.113144
5	MNB	0.875270	0.009834

Figure 29 : result after oversampling

In conclusion, we will use the logistic regression machine learning model , that has been trained using the dataset preprocessed by using textblob and lemmatization that has been applied to oversampling. The model is then used to deploy as the sentiment analysis system that can accurately predict the sentiment of comment in Malaysian Rojak language or mix language of malay and english. This can achieve our goal to design and implement a sentiment analysis system that can accurately analyze the sentiment of text written in out-of-vocabulary (OOV) Malaysia Rojak language. The final system is shown in figure 30.

Sentiment Analysis App

Enter text:

Analyze

Figure 30 : sentiment analysis system deployed

In this project, there are several limitations, which include dataset category limitation and lack of real time adaptability. Our dataset is scrapped from only tudung products from shopee, which means that the sentiment analysis system deployed can only predict the comment of tudung review. If comments from unrelated categories, like food and hotels, can lead to inaccurate predictions. Another limitation is that the sentiment analysis model faces challenges in adapting to the dynamic nature of language, especially when dealing with emerging words, phrases, and short forms. Language is dynamic and it continuously creates new words, phrases, and short forms. The sentiment analysis model is limited by its training data, and it may struggle to predict sentiments associated with newly coined terms or the latest linguistic trends, resulting in potential misinterpretations.

In the future, the dataset needs to be improved by adding more categories of product review into it, so that more category product reviews can be analyzed by using our sentiment analysis system. Next, the feedback feature may also need to be implemented in this system. This tool is used to analyze user-provided feedback and allows for a deeper understanding of the model's decision-making process and to improve the model's ability to predict the new words, phrases, and short forms.

6.0 References

1. *Benchmarking Machine Translated Sentiment Analysis for Arabic Tweets*. (2015, June 1). ACL Anthology. Retrieved December 22, 2023, from <https://aclanthology.org/N15-2010.pdf>
2. *BRCC and SentiBahasaRojak: The First Bahasa Rojak Corpus for Pretraining and Sentiment Analysis Dataset*. (2022, October 17). ACL Anthology. Retrieved December 21, 2023, from <https://aclanthology.org/2022.coling-1.389.pdf>
3. BS, S. (n.d.). (PDF) *An Interpretation of Lemmatization and Stemming in Natural Language Processing*. ResearchGate. Retrieved December 22, 2023, from https://www.researchgate.net/publication/348306833_An_Interpretation_of_Lemmatization_and_Stemming_in_Natural_Language_Processing
4. Çitak, E. (2023, July 28). *Data Preprocessing: Definition, Steps, And Requirements*. Dataconomy. Retrieved December 25, 2023, from <https://dataconomy.com/2023/07/28/data-preprocessing-steps-requirements/>
5. *A Comparative Study on TF-IDF Feature Weighting Method and its Analysis using Unstructured Dataset*. (n.d.). CEUR-WS. Retrieved December 25, 2023, from <https://ceur-ws.org/Vol-2870/paper10.pdf>
6. *Customer Sentiment Analysis: A crucial need in E-Commerce*. (2015, December 21). Decision Foundry. Retrieved December 22, 2023, from <https://www.decisionfoundry.com/business-intelligence-consulting/articles/customer-sentiment-analysis-in-e-commerce-data-initiatives/>
7. Dupont, M. (2023, July 21). *Manual Data Annotation vs. Augmented Annotation: Pros and Cons*. Labelvisor. Retrieved December 25, 2023, from <https://www.labelvisor.com/manual-data-annotation-vs-augmented-annotation-pros-and-cons/>

8. *THE DYNAMICS OF SELLER REPUTATION: EVIDENCE FROM EBAY*. (n.d.). NYU Stern. Retrieved December 22, 2023, from https://pages.stern.nyu.edu/~lcabral/workingpapers/CabralHortacsu_Aug08.pdf
9. Hutto, C. (n.d.). *vaderSentiment* · PyPI. PyPI. Retrieved December 25, 2023, from <https://pypi.org/project/vaderSentiment/>
10. *INTERNATIONAL YOUNG SCHOLARS JOURNAL OF LANGUAGES*. (n.d.). International Islamic University Malaysia. Retrieved December 25, 2023, from https://www.iium.edu.my/media/77652/Slang%20Language%20Use%20in%20Social%20Media%20Among%20Malaysian%20Youths_A%20Sociolinguistic%20Perspective.pdf
11. *Malaysia E-Commerce Market - Statistics, Share & Growth*. (n.d.). Mordor Intelligence. Retrieved December 22, 2023, from <https://www.mordorintelligence.com/industry-reports/malaysia-ecommerce-market>
12. McSwine, D. (2023, November 7). ., ., - YouTube. Retrieved December 25, 2023, from <https://link.springer.com/article/10.1007/s10588-018-9266-8>
13. Mitra, P. (n.d.). Research-paper recommender systems : a literature survey. Retrieved December 25, 2023, from <https://kops.uni-konstanz.de/server/api/core/bitstreams/8b886e4a-ea4b-4eae-bba1-19918f353170/content>
14. *A Mixed Malay–English Language COVID-19 Twitter Dataset: A Sentiment Analysis*. (n.d.). MDPI. Retrieved December 21, 2023, from <https://www.mdpi.com/2504-2289/7/2/61>
15. Nunzio, D. (n.d.). *A Linguistic Failure Analysis of Classification of Medical Publications: A Study on Stemming vs Lemmatization*. CEUR-WS. Retrieved December 22, 2023, from <https://ceur-ws.org/Vol-2253/paper45.pdf>
16. *Opinion mining and sentiment analysis*. (2007, October 16). Cornell CS. Retrieved December 25, 2023, from <https://www.cs.cornell.edu/home/llee/omsa/omsa.pdf>

17. Raileanu, G. (2023, March 3). *A Comprehensive Guide to 5-Star Surveys*. Retently. Retrieved December 25, 2023, from <https://www.retently.com/blog/5-star-survey/>
18. Roshan, R. (2023, September 20). *Comparative Analysis of TF-IDF and Hashing Vectorizer for Fake News Detection in Sindhi: A Machine Learning and Deep Learning Approach*. MDPI. Retrieved December 25, 2023, from <https://www.mdpi.com/2673-4591/46/1/5>
19. *Seeing stars: Exploiting class relationships for sentiment categorization with respect to rating scales*. (n.d.). Cornell Computer Science. Retrieved December 22, 2023, from <https://www.cs.cornell.edu/home/llee/papers/pang-lee-stars.pdf>
20. *Sentiment Analysis in Python: TextBlob vs Vader Sentiment vs Flair vs Building It From Scratch*. (n.d.). neptune.ai. Retrieved December 25, 2023, from <https://neptune.ai/blog/sentiment-analysis-python-textblob-vs-vader-vs-flair>
21. Shalunts, G., Backfried, G., & Commeignes, N. (2016). The impact of machine translation on sentiment analysis. *Data Analytics*, 63, 51-56. Retrieved December 21, 2023, from https://d1wqtxts1xzle7.cloudfront.net/86781204/download-libre.pdf?1654018383=&response-content-disposition=inline%3B+filename%3DThe_Impact_of_Machine_Translation_on_Sen.pdf&Expires=1703475125&Signature=DN2OLUw9pTk6UVBKsWIBfGSfRNNWcKtIt-n-ZACBIFT3G9vvXw46G1
22. *Shopee Scraper* · Apify. (n.d.). Apify. Retrieved December 22, 2023, from <https://apify.com/lukass/shopee-scraper>
23. Stepankova, B. (2023, August 22). *(PDF) Quality and Efficiency of Manual Annotation: Pre-annotation Bias*. ResearchGate. Retrieved December 25, 2023, from https://www.researchgate.net/publication/371606441_Quality_and_Efficiency_of_Manual_Annotation_Pre-annotation_Bias

24. *A Systematic Review of Emoji: Current Research and Future Perspectives*. (2019, October 15). NCBI. Retrieved December 25, 2023, from <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6803511/>
25. Tech Science Press. (2019, June 24). *Auto Annotation Technique*. A Novel Auto-Annotation Technique for Aspect Level Sentiment Analysis. Retrieved December 22, 2023, from https://www.researchgate.net/profile/Muhammad-Asif-160/publication/364352370_A_Novel_Auto-Annotation_Technique_for_Aspect_Level_Sentiment_Analysis/links/634e7bdb6e0d367d91a7ffbe/A-Novel-Auto-Annotation-Technique-for-Aspect-Level-Sentiment-Analysis.pdf
26. *Understanding Customer Experience*. (n.d.). Harvard Business Review. Retrieved December 21, 2023, from <https://hbr.org/2007/02/understanding-customer-experience>
27. *Vader Sentiment*. (2019, June 24). Retrieved December 25, 2023, from https://vadersentiment.readthedocs.io/en/latest/pages/about_the_scoring.html